	Marwadi University **Faculty of Technology** **Department of Information and Communication Technology**	
Digital Signal and Image Processing (01CT1513)	**Aim : Simulate cross correlation and autocorrelation on discrete time signals**	
Lab Experiment : 3	**Date: 31/07/2026**	**Enrolment No: 92400133125**

Aim : Simulate cross correlation and autocorrelation on discrete time signals.

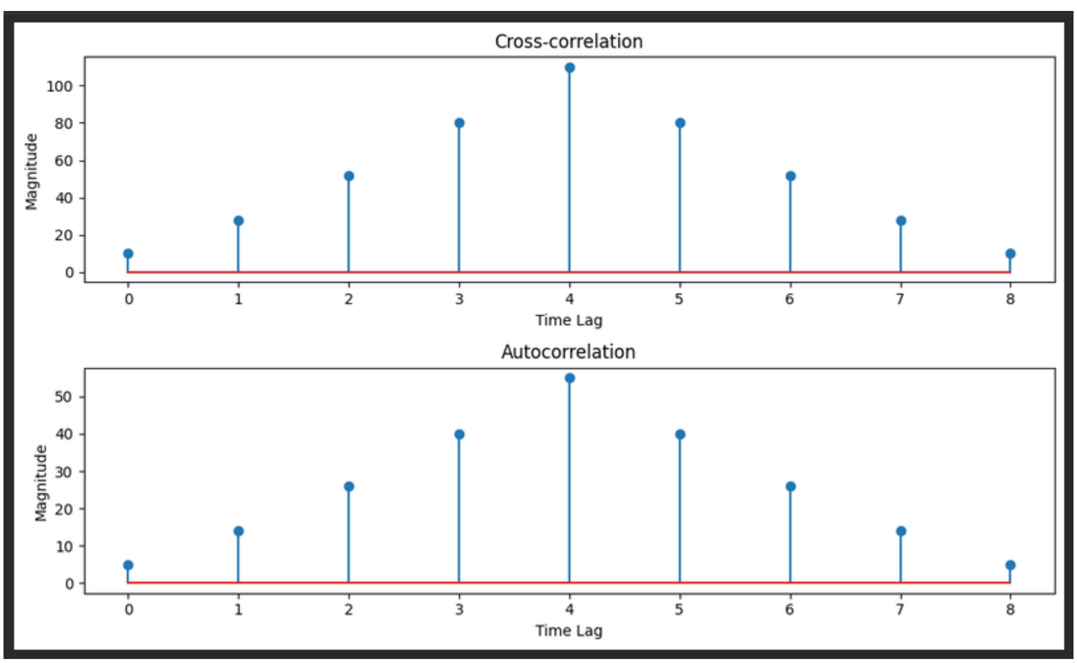
GitHub : https://github.com/mankumarbhalodiya/Digital-Signal-and-Image-Processing/tree/main/Experiment%203

Code :

```
import numpy as np

import matplotlib.pyplot as plt

def cross_correlation(signal1, signal2):

# Compute the cross-correlation

cross_corr = np.correlate(signal1, signal2, mode='full')

return cross_corr

def autocorrelation(signal):

# Compute the autocorrelation

auto_corr = np.correlate(signal, signal, mode='full')

return auto_corr

# Define the discrete-time signals

signal1 = np.array([1, 2, 3, 4, 5])

signal2 = np.array([2, 4, 6, 8, 10])

# Compute the cross-correlation

cross_corr = cross_correlation(signal1, signal2)

# Compute the autocorrelation

auto_corr = autocorrelation(signal1)

# Plot the cross-correlation and autocorrelation signals

plt.figure(figsize=(10, 6))

plt.subplot(2, 1, 1)
```

```
plt.stem(cross_corr)
plt.title('Cross-correlation')
plt.xlabel('Time Lag')
plt.ylabel('Magnitude')
plt.subplot(2, 1, 2)
plt.stem(auto_corr)
plt.title('Autocorrelation')
plt.xlabel('Time Lag')
plt.ylabel('Magnitude')
plt.tight_layout()
plt.show()
```

Output :



Conclusion :

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The implementation is able to compute the cross-correlation and autocorrelation of the provided signals. The cross-correlation provides information about the degree of similarity between two distinct signals at various time shifts, whereas the autocorrelation indicates how well the signal is similar to itself. As signal2 is scaled from signal1, there is a high peak for cross-correlation, and maximum value occurs at zero lag for autocorrelation.

Calculation :

Experiment 3 Calculation

$x(n) = [1, 2, 3, 4, 5]$ $\quad n = 0, 1, 2, 3, 4$

$y(n) = [2, 4, 6, 8, 10]$ $\quad n = 0, 1, 2, 3, 4$

→ Cross - Correlation

$$\gamma_{xy}(l) = \sum_{n=-\infty}^{\infty} x(n)\, y(n-l)$$

l	Shifted signal y(n-l)	Overlapping Sum Calculation	$\gamma_{xy}(l)$
-4	[0,0,0,0,2,4,6,8,10]	1x10	10
-3	[0,0,0,2,4,6,8,10,0]	1x8+2x10	28
-2	[0,0,2,4,6,8,10,0,0]	1x6 + 2x8 + 3x10	52
-1	[0,2,4,6,8,10,0,0,0]	1x4 + 2x6 + 3x8 + 4x10	80
0	[2,4,6,8,10,0,0,0,0]	1x2 + 2x4 + 3x6 + 4x8 + 5x10	110
1	[0,2,4,6,8,10,0,0,0]	2x2 + 3x4 + 4x6 + 5x8	80
2	[0,0,2,4,6,8,10,0,0]	3x2 + 4x4 + 5x6	52
3	[0,0,0,2,4,6,8,10,0]	4x2 + 5x4	28
4	[0,0,0,0,2,4,6,8,10]	5x2	10

Cross - Correlation = [10, 28, 52, 80, 110, 80, 52, 28, 10]

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Auto - correlation

$$r_{xx}(l) = \sum_{n=-\infty}^{\infty} x(n)\, x(n-l)$$

l	Calculation	$r_{xx}(l)$
-4	1x5	5
-3	1x4 + 2x5	14
-2	1x3 + 2x4 + 3x5	26
-1	1x2 + 2x3 + 3x4 + 4x5	40
0	$1^2 + 2^2 + 3^2 + 4^2 + 5^2$	55
1	2x1 + 3x2 + 4x3 + 5x4	40
2	3x1 + 4x2 + 5x3	26
3	4x1 + 5x2	14
4	5x1	5

auto-correlation: [5, 14, 26, 40, 55, 40, 26, 14, 5]